CLINISCAN PROJECT — BATCH B13

Milestone 2

Model Development & Baseline Training

Chest X-ray Lung Abnormality Detection System

Author	**Dataset**	**Models**	**Timeline**
Lucky Soni	VinBigData CXR (Kaggle)	ResNet50 + YOLOv8s	Weeks 3 – 4

MILESTONE SCOPE

✔ Multi-Label Classification Pipeline	✔ ResNet50 Transfer Learning
✔ PyTorch Dataset & DataLoader	✔ Train / Val / Test Split
✔ YOLO Annotation Conversion	✔ YOLOv8s Detection Training
✔ Evaluation Metrics (AUC, F1, mAP)	✔ Detection Visualizations & Plots

Test AUC	Test F1	mAP50	Images	Classes
0.884	**0.297**	**0.162**	**5,000**	**14**
Classification	Classification	Detection	Training Data	Pathologies

PART A

PROJECT OVERVIEW & OBJECTIVE

1. Milestone Objective

Milestone 2 focuses on implementing and training two distinct deep learning models on the preprocessed VinBigData Chest X-ray dataset. The primary goal is to establish a solid baseline for both classification and object detection tasks — forming the foundation for hyperparameter optimization in Milestone 3.

Two models were developed:

- ■ **Classification Model** — Predicts *which* abnormalities are present using multi-label classification (ResNet50).
- ■ **Detection Model** — Predicts *where* abnormalities are located using bounding boxes with class labels (YOLOv8s).

2. Dataset Summary

Component	Details
Total Images Used	5,000 preprocessed PNG images (random subset)
Image Dimensions	224 x 224 pixels — converted from DICOM via Kaggle pipeline
Annotation Format	CSV bounding boxes (original) → YOLO normalized format (converted)
Number of Classes	14 thoracic abnormalities + 1 normal class (No finding)
Train / Val / Test Split	4,000 / 500 / 500 images — 80% / 10% / 10%
Annotation Source	train.csv — VinBigData Kaggle competition
Original DICOM Resolution	Variable — up to 3,025 x 3,384 pixels
Total Bounding Box Annotations	11,860 across 5,000 images (after No finding removal)

3. Class Distribution

The dataset exhibits significant class imbalance — a typical characteristic of medical imaging datasets. The most prevalent finding is 'No finding' (normal scans), while rare pathologies such as Pneumothorax appear infrequently. This imbalance was a key consideration during training and evaluation metric selection.

Class	Instances	Frequency	Category
No finding	10,647	47.3%	Normal
Aortic enlargement	2,315	10.3%	Common

Cardiomegaly	1,733	7.7%	Common
Pleural thickening	1,576	7.0%	Common
Pulmonary fibrosis	1,528	6.8%	Common
Nodule/Mass	884	3.9%	Moderate
Pleural effusion	846	3.8%	Moderate
Lung Opacity	843	3.7%	Moderate
Other lesion	733	3.3%	Moderate
Infiltration	403	1.8%	Rare
ILD	364	1.6%	Rare
Calcification	308	1.4%	Rare
Consolidation	149	0.7%	Rare
Atelectasis	105	0.5%	Rare
Pneumothorax	73	0.3%	Very Rare

PART B

DATA PREPARATION PIPELINE

1. Multi-Label Matrix Construction

Since a single chest X-ray can simultaneously contain multiple abnormalities, the classification task is framed as a multi-label problem. A binary label matrix was constructed where each row represents an image and each column represents a pathology class.

- Filtered out 'No finding' entries (class_id = 14) before constructing the findings matrix.
- Used pandas.crosstab() to pivot image-level annotations into a binary presence matrix.
- Re-indexed against all 5,000 image IDs, filling missing entries with 0 (no abnormality detected).
- Result: a 5,000 × 14 binary matrix — one flag per pathology per image.

Metric	Value
Images with No Finding (all zeros)	3,549 images (70.98%)
Images with exactly 1 finding	133 images (2.66%)
Images with 2 or more findings	1,318 images (26.36%)
Final matrix shape	5,000 rows × 14 columns

2. Train / Validation / Test Split

The dataset was partitioned into three non-overlapping subsets using sklearn's train_test_split with a fixed random seed (42) for reproducibility. Splitting was performed at the image level to prevent data leakage between subsets.

Split	Images	Proportion	Purpose
Train	4,000	80%	Model parameter optimization via gradient updates
Validation	500	10%	Hyperparameter tuning and early stopping decisions
Test	500	10%	Final unbiased performance evaluation — held out

3. YOLO Annotation Conversion

Bounding box annotations from train.csv were converted to YOLO normalized format. The key challenge was accurately scaling from original DICOM resolution (variable, up to 3,384 pixels) to the processed 224×224 images.

- DICOM metadata (Rows, Columns) extracted per image using pydicom with stop_before_pixels=True — metadata-only read completes in under 5 minutes versus 1.5 hours for full pixel extraction.

• Per-image scale factors computed: scale_x = 224 / orig_w, scale_y = 224 / orig_h.

• YOLO coordinates normalized to [0.0, 1.0] range and clipped to prevent out-of-bounds values.

• Class ID 14 (No finding) entries excluded — these produce empty .txt label files (background images).

• Total annotations successfully converted: 11,860 bounding boxes across 5,000 label files.

```
center_x = ((x_min + x_max) / 2) / orig_w * (224 / orig_w) → normalized to [0,1]
center_y = ((y_min + y_max) / 2) / orig_h * (224 / orig_h) → normalized to [0,1]
width = (x_max - x_min) / orig_w → normalized width
height = (y_max - y_min) / orig_h → normalized height
```

PART C

CLASSIFICATION MODEL — ResNet50

1. Model Architecture

ResNet50 (Deep Residual Network, 50 layers) was selected as the classification backbone. Pretrained ImageNet weights were loaded via PyTorch torchvision to leverage transfer learning — enabling the model to extract meaningful visual features from a relatively small medical imaging dataset.

Component	Specification
Base Model	ResNet50 — pretrained on ImageNet (torchvision.models.resnet50)
Input Shape	(batch_size, 3, 224, 224) — RGB image tensor
Backbone Output	2,048-dimensional feature vector from Global Average Pooling
Modified Head	Linear(2048 → 14) — custom classification layer for 14 pathologies
Output Shape	(batch_size, 14) — raw logits, one per pathology class
Activation	Sigmoid applied post-output for multi-label probability estimation
Parameters	~25.6 million total trainable parameters
Pretrained	Yes — ImageNet weights (DEFAULT) transferred to backbone

2. Training Configuration

Hyperparameter	Value	Justification
Loss Function	BCEWithLogitsLoss	Standard loss for multi-label binary classification — numerically stable
Optimizer	Adam	Adaptive learning rate — robust for medical imaging tasks
Learning Rate	0.001	Standard starting LR for fine-tuning pretrained models
Batch Size	16	Memory-efficient for CPU baseline training
Epochs	10	Baseline run — extended training planned for Milestone 3
LR Scheduler	ReduceLROnPlateau	Halves LR when validation loss stagnates for 2 epochs
Scheduler Factor	0.5	50% LR reduction per plateau event
Augmentation	HFlip + Rotation(10°) + ColorJitter	Safe medical augmentations — no vertical flips or perspective warping

Normalization	ImageNet mean/std	Mean=[0.485, 0.456, 0.406], Std=[0.229, 0.224, 0.225]
Device	CPU	GPU not available for baseline — GPU planned for Milestone 3

3. Epoch-by-Epoch Training Log

Epoch	Train Loss	Val Loss	Val F1	Val AUC	Status
1	0.2084	0.2006	0.0830	0.7939	—
2	0.1834	0.1846	0.1117	0.8096	✔ Best saved
3	0.1680	0.1881	0.0905	0.8352	—
4	0.1647	0.2009	0.0888	0.8524	—
5	0.1608	0.1798	0.1401	0.8230	✔ Best saved
6	0.1524	0.1671	0.1338	0.8635	—
7	0.1496	0.1695	0.1755	0.8464	✔ Best saved
8	0.1493	0.1879	0.1381	0.8433	—
9	0.1419	0.1563	0.2088	0.8752	✔ Best saved
10	**0.1405**	**0.1744**	**0.2704**	**0.8644**	**✔ Final best**

Highlighted row = best model checkpoint saved to best_model.pth

4. Final Test Set Evaluation

Metric	Score	Interpretation
Test Loss	0.1803	Low generalization loss — model not overfitting to training data
Test F1 (macro)	0.2967	Baseline score — class imbalance on rare classes limits macro average
Test AUC (macro)	0.8844	Strong discriminative ability — robust ranking across all 14 classes

5. Per-Class F1 Score Analysis

Class	F1 Score	Sample Count	Analysis
Aortic enlargement	0.8143	2,315	High — most frequent pathology with clear visual features
Cardiomegaly	0.8000	1,733	High — distinct enlarged heart silhouette

Lung Opacity	0.4602	843	Moderate — diffuse patterns challenging to distinguish
Pleural thickening	0.5198	1,576	Moderate — linear features learnable at 224x224
Pleural effusion	0.5000	846	Moderate — blunted costophrenic angles detected
ILD	0.4000	364	Moderate — complex reticular patterns
Pulmonary fibrosis	0.4125	1,528	Moderate — fibrotic changes captured partially
Infiltration	0.0769	403	Low — visually similar to opacity, overlap confusion
Other lesion	0.1695	733	Low — heterogeneous class, no distinct visual pattern
Atelectasis	0.0000	105	Zero — insufficient samples, model unable to learn
Calcification	0.0000	308	Zero — small focal lesions below detection threshold at 224px
Consolidation	0.0000	149	Zero — very few training examples after split
Nodule/Mass	0.0000	884	Zero — subtle lesions require higher resolution and more epochs
Pneumothorax	0.0000	73	Zero — extremely rare, only 73 instances in full dataset

PART D

OBJECT DETECTION MODEL — YOLOv8s

1. Model Selection & Architecture

YOLOv8 Small (YOLOv8s) was selected as the detection model for its balance of training speed and detection accuracy. Pretrained COCO weights were fine-tuned on the VinBigData dataset. YOLOv8 is a single-stage detector that simultaneously regresses bounding box coordinates and predicts class probabilities.

Component	Specification
Model Variant	YOLOv8s (Small) — yolov8s.pt pretrained on COCO
Architecture	130 layers — CSP backbone + PANet neck + Detect head
Parameters	11.1 million (11,141,018 total) — 28.7 GFLOPs
Weight Transfer	349 / 355 layers successfully transferred from pretrained COCO weights
Output Classes	14 (overriding COCO default of 80 classes)
Detection Type	Single-stage — simultaneous box regression and classification
Augmentation	Built-in Albumentations: Blur, MedianBlur, CLAHE, ToGray

2. Training Configuration

Parameter	Value	Notes
Base Weights	yolov8s.pt	COCO pretrained — transfer learning
Epochs	10	Baseline — GPU would allow 50+ epochs
Image Size	224 × 224	Matches preprocessed image dimensions
Batch Size	16	Memory-efficient for CPU training
Optimizer	AdamW (auto)	lr=0.000556, momentum=0.9, decay=0.0005
Box Loss	7.5	YOLOv8 default bounding box regression weight
Class Loss	0.5	YOLOv8 default classification weight
DFL Loss	1.5	Distribution Focal Loss — fine-grained box prediction
Patience	5 epochs	Early stopping if mAP does not improve
Device	CPU	1.278 hours for 10 epochs on AMD Ryzen 7 PRO 5850U

3. Dataset Split for Detection

Split	Images	Label Files	Background Images	Notes
Train	4,001	3,999	2,831	Background = No finding images
Val	1,001	1,001	720	Used for mAP evaluation per epoch

4. Epoch Training Progress

Epoch	Box Loss	Cls Loss	DFL Loss	mAP50	mAP50-95
1	2.605	4.780	1.891	0.0563	0.0232
2	2.133	3.031	1.553	0.0794	0.0337
3	2.064	2.732	1.500	0.0860	0.0376
4	2.092	2.392	1.557	0.1230	0.0546
5	1.997	2.313	1.467	0.1160	0.0517
6	1.950	2.154	1.441	0.1150	0.0522
7	1.880	2.080	1.402	0.1360	0.0569
8	1.866	1.948	1.394	0.1380	0.0622
9	1.838	1.938	1.384	0.1580	0.0727
10	**1.815**	**1.894**	**1.355**	**0.1620**	**0.0739**

Consistent loss reduction across all 10 epochs. Highlighted = best model (best.pt saved).

5. Per-Class Detection Results — Best Model Validation

Class	Images	Instances	Precision	Recall	mAP50	mAP50-95
Aortic enlargement	203	474	0.464	0.565	0.551	0.262
Atelectasis	13	23	1.000	0.000	0.132	0.040
Calcification	31	58	1.000	0.000	0.001	0.000
Cardiomegaly	143	339	0.452	0.640	0.585	0.364
Consolidation	24	33	0.311	0.121	0.077	0.027
ILD	34	102	0.309	0.245	0.198	0.081
Infiltration	38	67	0.466	0.149	0.153	0.062
Lung Opacity	83	154	0.164	0.143	0.091	0.032
Nodule/Mass	54	145	0.000	0.000	0.008	0.003
Other lesion	66	138	0.506	0.051	0.042	0.012

Pleural effusion	61	150	0.221	0.447	0.289	0.112
Pleural thickening	122	278	0.151	0.108	0.055	0.013
Pneumothorax	4	8	1.000	0.000	0.005	0.001
Pulmonary fibrosis	94	248	0.184	0.105	0.076	0.027
ALL (overall)	**1001**	**2217**	**0.445**	**0.184**	**0.162**	**0.074**

Highlighted row = overall model performance across all 14 classes.

PART E

ANALYSIS, CHALLENGES & OBSERVATIONS

1. Classification Model — Strengths & Limitations

STRENGTHS	LIMITATIONS
✔ Training loss decreased consistently from 0.208 to 0.140 across 10 epochs. ✔ Test AUC of 0.884 — strong discriminative ability across all 14 classes. ✔ Common pathologies (Aortic enlargement F1=0.81, Cardiomegaly F1=0.80) learned effectively. ✔ Transfer learning from ImageNet enabled fast convergence within 10 epochs.	✘ Rare classes (Pneumothorax=73, Atelectasis=105) — zero F1 due to insufficient samples. ✘ Macro F1 of 0.297 dragged down by zero-F1 rare classes — overall metric misleading. ✘ No class-weighted loss applied at baseline — all classes treated equally. ✘ CPU training limited to 10 epochs — extended training required for convergence.

2. Detection Model — Strengths & Limitations

STRENGTHS	LIMITATIONS
✔ Consistent mAP50 improvement from 0.056 → 0.162 across 10 epochs. ✔ Cardiomegaly (mAP50=0.585) and Aortic enlargement (mAP50=0.551) — strong localization. ✔ All three losses (box, cls, dfl) decreased monotonically — stable training. ✔ 349/355 pretrained layers transferred successfully from COCO.	✘ Rare classes (Pneumothorax: 4 val images) — insufficient instances for bounding box learning. ✘ Image size 224x224 (below recommended 640x640) — small lesions below detection resolution. ✘ CPU training restricted to 10 epochs — 50+ epochs required for full convergence. ✘ High proportion of background images (2,831 train) may dilute detection gradient signal.

3. Technical Challenges Encountered

Challenge	Description	Resolution
DICOM Coordinate Scaling	Original DICOM images varied up to 3,384px. Fixed-size division was incorrect — all normalized coordinates became 1.0.	Extracted per-image dimensions from DICOM metadata using pydicom stop_before_pixels=True on Kaggle.

Kaggle Session Expiry	Kaggle /working/ directory is temporary. Generated YOLO labels were lost between sessions.	Metadata-only DICOM reading reduced re-run from 1.5 hours to under 5 minutes.
OMP Runtime Conflict	Python 3.13 Anaconda base had duplicate libiomp5md.dll — PyTorch kernel crashed on every run.	Set KMP_DUPLICATE_LIB_OK=TRUE as first line before all imports in every notebook.
CSV vs Folder Mismatch	subset_train.csv was manually created and contained 3,330 image IDs not present in processed folder.	Filtered train.csv against actual folder image IDs — created my_5000_labels.csv with exact 5,000 match.

PART F

PROJECT STRUCTURE & DELIVERABLES

1. Project Folder Structure

```
B13-CliniScan/
■■■ data/
■ ■■■ raw/
■ ■ ■■■ train/ (original DICOM files — not tracked in git)
■ ■ ■■■ train.csv (full VinBigData annotation file)
■ ■ ■■■ my_5000_labels.csv (filtered annotations for 5,000 images)
■ ■ ■■■ multilabel_matrix.csv (5000 x 14 binary label matrix)
■ ■ ■■■ train_split.csv (4,000 training image IDs + labels)
■ ■ ■■■ val_split.csv (500 validation image IDs + labels)
■ ■ ■■■ test_split.csv (500 test image IDs + labels)
■ ■■■ processed/
■ ■ ■■■ labels_yolo/ (5,000 YOLO .txt annotation files)
■ ■ ■■■ sample_images.png (EDA visualization — 6 sample X-rays)
■ ■ ■■■ class_distribution.png (class imbalance bar chart)
■ ■ ■■■ training_curves.png (loss + F1 curves for classification)
■ ■■■ yolo_dataset/
■ ■ ■■■ train/images/ (4,001 training images for YOLOv8)
■ ■ ■■■ train/labels/ (3,999 YOLO label files — train)
■ ■ ■■■ val/images/ (1,001 validation images)
■ ■ ■■■ val/labels/ (1,001 YOLO label files — val)
■ ■ ■■■ dataset.yaml (YOLOv8 configuration file)
■ ■■■ detection_results/
■ ■ ■■■ map_results_table.csv (per-class mAP50 / mAP50-95 results)
■ ■ ■■■ map50_per_class.png (mAP50 bar chart per class)
■ ■ ■■■ precision_recall_plot.png (precision vs recall scatter plot)
■ ■ ■■■ [20 detection PNG images] (sample bounding box visualizations)
■ ■■■ best_model.pth (ResNet50 best classification weights)
■■■ docs/
■ ■■■ Dataset_Description.pdf (Milestone 1 dataset documentation)
■ ■■■ Milestone1_Report.pdf
■ ■■■ Milestone2_Report.pdf (this document)
■■■ runs/
■ ■■■ detect/data/yolo_runs/cliniscan_detection/
■ ■■■ weights/best.pt (YOLOv8s best detection weights)
■ ■■■ weights/last.pt (YOLOv8s last epoch weights)
■ ■■■ [training plots & logs]
■■■ src/
■ ■■■ milestone2_classification.ipynb (classification training notebook)
■ ■■■ milestone2_detection.ipynb (YOLOv8 detection training notebook)
■ ■■■ annotation.ipynb (YOLO annotation conversion notebook)
■ ■■■ dicom_to_png.py (Milestone 1 — DICOM conversion script)
■ ■■■ preprocessing.py (Milestone 1 — image preprocessing)
■ ■■■ annotation_conversion.py (Milestone 1 — annotation conversion)
■■■ .gitignore
```

```
■■■ LICENSE
■■■ README.md
■■■ requirements.txt
```

2. Milestone 2 Deliverables Checklist

✔ Multi-label binary matrix constructed from train.csv (5,000 × 14)
✔ Train / Val / Test split completed — 80% / 10% / 10%
✔ PyTorch ChestXrayDataset class and DataLoader pipeline implemented
✔ ResNet50 classification model trained — 10 epochs, best model saved
✔ Training loss and validation F1 / AUC tracked and plotted
✔ Test set evaluation: AUC = 0.884, F1 = 0.297, Loss = 0.180
✔ Per-class F1 scores computed and documented for all 14 classes
✔ YOLO annotations correctly scaled from original DICOM resolution per image
✔ YOLOv8s detection model trained — 10 epochs, best.pt and last.pt saved
✔ Detection metrics: mAP50 = 0.162, mAP50-95 = 0.074 reported
✔ Per-class bounding box detection results documented
✔ 20 detection visualization images with predicted bounding boxes generated
✔ mAP50 per-class bar chart and Precision-Recall scatter plot saved
✔ mAP results table exported to CSV
✔ Code organized into modular Jupyter notebooks in src/ folder
✔ All outputs and results committed to GitHub repository (B13-CliniScan)

3. Final Evaluation Metrics Summary

Model	Metric	Score	Status
ResNet50 Classification	Test AUC (macro)	0.8844	✔ Strong baseline
ResNet50 Classification	Test F1 (macro)	0.2967	✔ Imbalance limited — expected
ResNet50 Classification	Test Loss	0.1803	✔ Good generalization
YOLOv8s Detection	Overall mAP50	0.1620	✔ Baseline established
YOLOv8s Detection	Overall mAP50-95	0.0741	✔ Room for improvement
YOLOv8s Detection	Precision	0.4450	✔ Moderate precision
YOLOv8s Detection	Recall	0.1840	✔ Low — rare class issue

PART G

CONCLUSION & NEXT STEPS

Conclusion

Milestone 2 successfully delivered two fully functional deep learning pipelines for chest X-ray abnormality analysis. Both the classification model (ResNet50) and the detection model (YOLOv8s) were trained from scratch, evaluated with appropriate metrics, and their outputs documented with visualizations and performance tables.

The classification model achieved a test AUC of 0.884 — a strong result for a 10-epoch CPU-trained baseline. The detection model established a baseline mAP50 of 0.162 with top pathologies (Cardiomegaly: 0.585, Aortic enlargement: 0.551) demonstrating that the model successfully learned to localize the most common chest abnormalities.

The primary challenge across both models was class imbalance — an inherent property of medical imaging datasets. Rare pathologies with fewer than 100 training samples could not be adequately learned within a baseline 10-epoch run. This is expected at this stage and will be systematically addressed in Milestone 3.

Planned Improvements for Milestone 3

Area	Planned Action	Expected Impact
Class Imbalance	Apply inverse-frequency weighted BCEWithLogitsLoss	Improved F1 for rare pathologies
Training Duration	Extend to 30 epochs (classification), 50 epochs (detection)	Better convergence and generalization
Image Resolution	Train YOLOv8 at imgsz=640 for improved small lesion detection	Higher mAP50 for Nodule/Mass, Calcification
Hyperparameter Search	Grid search: LR ∊ {0.01, 0.001, 0.0001}, batch ∊ {16, 32}	Identify optimal training configuration
Advanced Augmentation	Albumentations pipeline: ElasticTransform, GridDistortion (safe)	Improved model robustness
Interpretability	Implement Grad-CAM heatmaps for classification model	Clinical interpretability of predictions
Model Upgrade	Evaluate YOLOv8m (medium) vs YOLOv8s on detection task	Potential mAP improvement with larger model
GPU Training	Leverage GPU compute for full training runs	10x+ training speed — enables more epochs